MIL-P-63061A (AR)
17 May 1978
SUPERSEDES
MIL-M-63061
1 June 1976

MILITARY SPECIFICATION

PROJECTILE, 155MM, M692/M731

PLUG, HOUSING FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Plug, Housing for the munition used in Projectiles, 155MM, M692 and M731.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

MIL-M-48245 - Molding Compound, Housing and Plug

STANDARDS

FEDERAL

FED-STD-595 - Colors

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9298401 - Plug, Housing

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-695 (69) - Compressive Properties of Rigid Plastics

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphis, PA 19103.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Plug, housing. - The plug shall comply with all requirements specified on Drawing (Dwg.) 9298401, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. Cleaning methods used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

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3.4 Void criteria for plug, housing. - Voids, breakouts, inperfections are allowed on the flat surface between the largest outside diameter and the large cavity to maximum dimensions of 3/8 inch across by 1/8 inch deep. Similar imperfections are allowed on the remainder of the plug to maximum dimensions of 1/8 inch across by 1/16 inch deep except on the arc face side where the maximum dimensions allowed shall be 1/32 inch across by 1/32 inch deep. Comparison photographs or visual standards as approved by the Contracting Officer may be used to establish visual accept/reject criteria for voids. Patching of voids in excess of the specified criteria is not permitted.

3.5 Compressive strength of molding compound. - The molding compound for the Plug, Housing shall exhibit a compressive strength of 29,000 to 38,200 psi.

3.6 Color. - The color of the plug, housing shall be within the range of colors bounded by color numbers 34079 and 34102, inclusive of FED-STD-595.

3.7 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawings	Quantity
Plug, Housing	9298401	25
Molding Compound	4.5.1	Two (2) test bars from each batch of molding compound used to mold the plug

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63061A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Plug, Housing				See Below
					NEXT HIGHER ASSEMBLY
					—
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Plug, Housing (See Dwg 9298401)				
	Examination for Defects	25		3.2	4.4.2.1
	Compressive Strength	2*		3.5	4.5.1.3
	Molding Compound	-		3.1	Certification**

NOTES:
*Two test bars from each batch of molding compound used to mold the plug
**Certification of compliance that the molding compound used to mold the plugs complies with all requirements specified in MIL-U-48245.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Plug, Housing shall contain molding compound of one lot interfix number from one manufacturer complying with all requirements of MIL-M-48245.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63061A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.1	Plug, Housing (See Note)				9298401
					NEXT HIGHER ASSEMBLY
					—
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Web thickness at bottom of cavity, min		0.40%	3.2	Gage
102	Diameter over ridges, maximum (2 places)		0.40%	3.2	Gage
103	Voids in excess of specified criteria		0.40%	3.4	Visual
104	Any protrusion on surface of cavity		0.40%	3.2	Visual
105	Resin/hardener weight ratio of molding compound		NA	3.2	4.5.1.2
106	Compressive strength		NA	3.5	4.5.1.3
107	Molding compound shelf life expired	NA	NA	3.2	Visual
108	Color incorrect		0.40%	3.6	Visual
109	Temperature control during cure and cure time		Note 1	3.2	Note 1
Minor					
201	Arc face contour		0.65%	3.2	Gage
202	Outside diameter, maximum		0.65%	3.2	Gage
203	Web thickness at bottom of cavity, max		0.65%	3.2	Gage
204	Surface preparation on applicable surfaces missing or incomplete		0.65%	3.2	Visual
205	Presence of flash		0.65%	3.2	Visual
206	Evidence of poor workmanship		1.0%	3.3	Visual
NOTE 1:	Cure temperature and cure time shall be monitored for each molding compound batch of plugs.				

NOTES: All plugs subjected to these examinations and tests shall be handled with suitable hand coverings to prevent comtamination of the "grit blasted" surface.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63061A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Preservation, Packaging and Packing				Section 5
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Minor					
201	Packaging improper		0.65%	3.3	5.1.1
202	Evidence of poor workmanship		1.00%	3.3	5.1.1, 5.1.2

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Tables.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/ Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

*4.5.1 Quality controls for each molding compound batch for molding plugs. -

*4.5.1.1 Molding compound batch definition. - A batch of molding compound is herein defined as that quantity of compound mixed at one time using base resin and hardener of one lot number from one manufacturer to mold plugs in one continuous unchanged process. When a continuous-type mixing process is used, a batch is that quantity of molding compound mixed and dispensed in one continuous unchanged process during a one (1) hour period, or portion thereof, using base resin and hardener of one lot number from one manufacturer.

4.5.1.2 Resin/hardener weight ratio control. - At the beginning of produciton and at least once each four hours of produciton, or portion thereof, the inspector shall draw two test shots with the mixer set to expel the hardener and resin separately. The hardener and resin of each shot shall be weighed individually to verify that the ratio is between 1.8/1 and 2.2/1. If either shot fails to meet the requirement, all items filled since the last acceptable check shall be rejected. Further, the mixer shall be corrected and shall pass the above test before being returned to use.

4.5.1.3 Mixing and cure control. - One (1) test bar from each batch of molding compound (See 4.5.1.1) shall be molded at the same time the batch is used to mold plugs. Vibration and evacuation as necessary may be used to mold the bar. The mix ratio limits shall be 2.0/1.0 + 5% (1.8/1 to 2.2/1). The test bar shall be processed with the

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plugs molded from the batch of molding compound which the bar represents, i.e., the bar shall be cured in the same ovens and subjected to the same temperatures for the same time periods as the items it represents. After completion of the cure, the bar shall be tested for ultimate compressive strength, as specified in ASTM-D-695 (69) except that the specimen dimensions shall be 0.5 inches by 0.5 inches by 1.0 inches or 0.5 diameter X 1.0 inches long. Tolerances on specimen dimensions are ± 0.01 inches. The ends shall be flat and parallel to within .003 inches, and shall be perpendicular to the long axis of the specimen. Speed of testing shall be as specified in the ASTM. If the results of this test are outside the limits specified in 3.5, the items represented by the bar shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - The plug, housing shall be preserved and packaged in accordance with MIL-STD-1169. The requirement for waterproof packages shall apply. The plugs shall be packaged in cardboard tubes. Packaging shall be satisfactory to the Contracting Officer.

5.2 Packing. - The plug, housing, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

*5.3 Marking. - Marking of the box for the plug, housing shall be in accordance with MIL-STD-1169.

5.4 Shipping. - When components from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - The plugs covered by this specification are intended for use in the munition for Projectiles, 155MM, M692 and M731.

6.2 Ordering data. - See MIL-A-48078.

*6.3 Submission of designs for approval. - See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

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Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Toxicity of molding compound. - When handling, mixing or curing the molding compound or its constituent, suitable precaustions must be taken, including proper ventilation, as some of the constituents and compounds are toxic and/or exude toxic fumes when mixed or heated. Use adequate ventilation. Avoid contact with skin, eyes and clothing and prolonged or repeated breathing of fumes.

*6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading U.S Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (addition, modifications, corrections, deletions) from the previous specification were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous specification.

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